JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SREE ANNAMARR AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

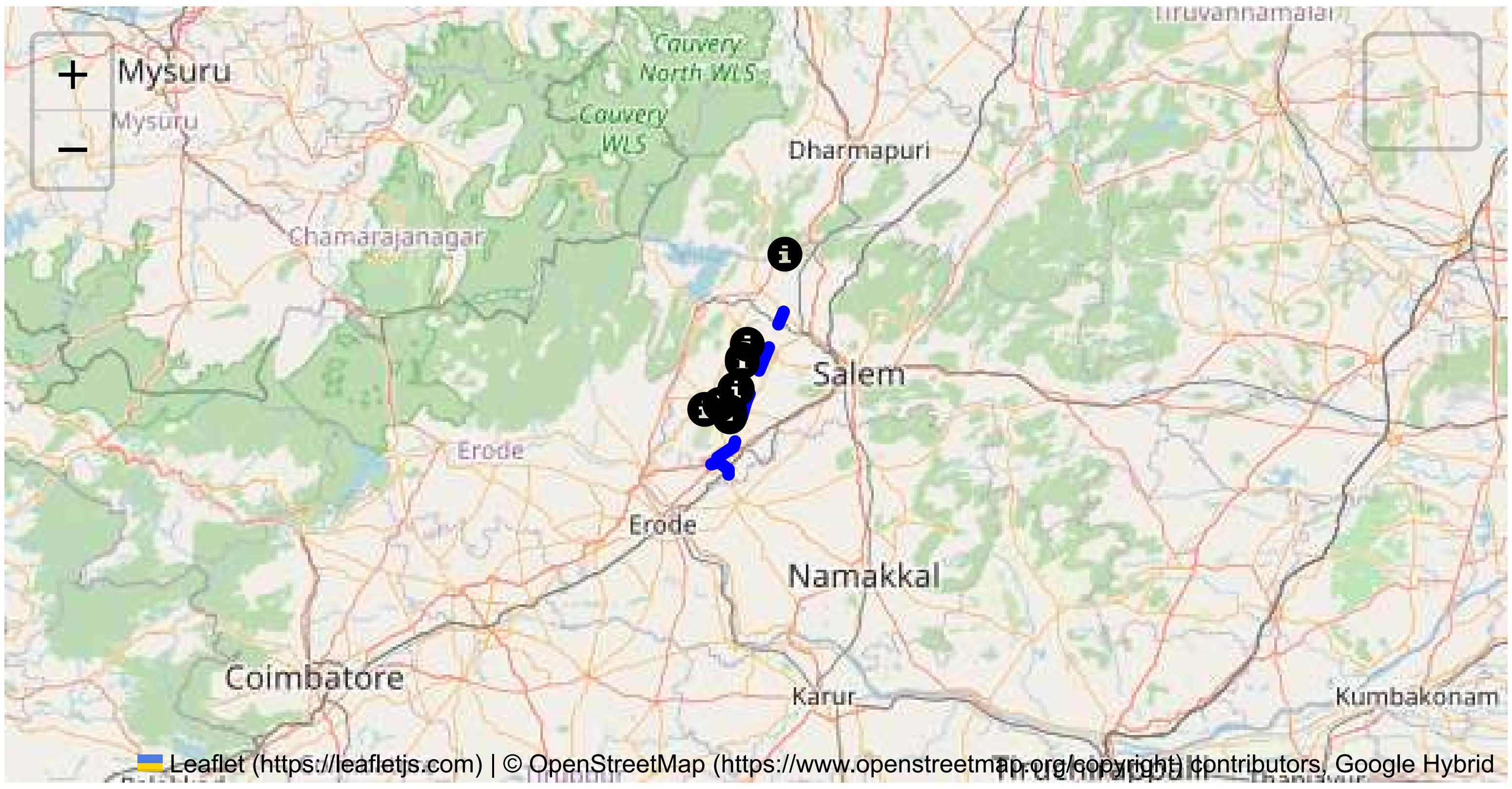
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

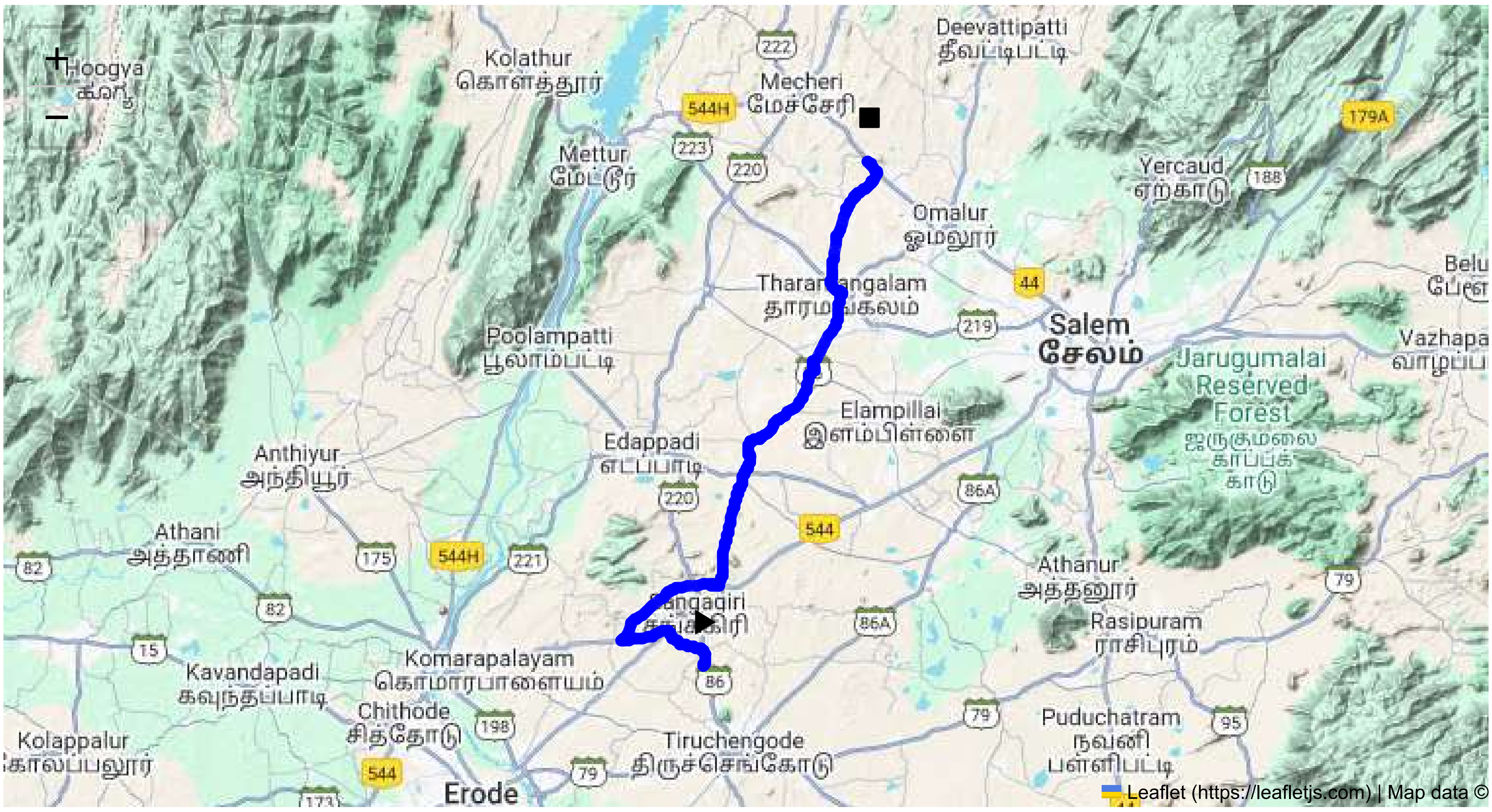
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 55.52 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.7 hours
Start: (11.4381, 77.8734)
End: (11.78546, 77.98946)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to QXPQ+6R2, Omalur, spans approximately 55.52 kilometers through Tamil Nadu. It includes several key waypoints such as Sankari RS - Erode Privu Rd, Bhavani Main Road in Sankari West and Padaiveedu, SH 86 through Ayveli, Erumaipatti, and Koranampatti. The direct path primarily takes SH 86, a significant state highway known for its rural expanses.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate. The region can face heavy rainfall during the monsoon season, primarily between June and September. This can lead to waterlogging and poor road visibility, potentially hazardous for truck drivers. In the summer months (March to May), high temperatures could affect driver concentration and vehicle performance.

3. Analysis of Traffic Patterns

Traffic patterns in these areas usually demonstrate light to moderate congestion, with potential increases during the mornings (8 AM to 10 AM) and evenings (5 PM to 7 PM). Sankari and Erumaipatti might experience more congestion due to local activities. Main roads and intersections close to urbanized towns could be other congestion points.

4. Assessment of Road Quality and Infrastructure

Road quality varies across this route. SH 86 is generally well-maintained but may include stretches of uneven surfaces and potholes, especially post-monsoon. Rural roads leading into highway segments might lack appropriate shoulder width and signage.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies or roadblocks, vehicles can take the Salem Bypass to re-route more efficiently towards Omalur via NH 44, although this adds distance. Keeping local secondary roads in mind can facilitate quick shifts in the route.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu mandates specific permissions for transporting hazardous materials. Vehicles must abide by rules regarding container specifications, emergency response equipment, and driver qualifications. Awareness of specific detours or time restrictions (like night curfews in certain areas) is also crucial.

7. Overview of Historical Incidents

While detailed incident data may be non-public, incidents involving heavy vehicles often occur due to overloading, speeding, or poor road conditions. The region hasn’t been notably incident-prone for hazardous materials, but vigilance remains instrumental.

8. Environmental Considerations and Sensitive Areas

The route passes close to agricultural lands, requiring vigilant driving to avoid contamination risks. Minimizing emissions and noise pollution is crucial, particularly through inhabited segments.

9. Analysis of Communication Coverage

While mobile network coverage is largely reliable along SH 86, rural transitions may experience short-lived dead zones, especially in less populated regions. Keeping alternative communication means, like satellite phones or radio, is advisable.

10. Estimated Emergency Response Times

Emergency services might take 20 to 40 minutes, depending on proximity to nearby townships like Sankari or Omalur. Planning with knowledge of local emergency contacts speeds up responsiveness during crises.

11. Overall Summary of Risk Assessment

This route poses moderate risks largely manageable with appropriate precautions. Weather conditions, traffic and road integrity are primary concerns. Adherence to safety regulations and maintaining alternative pathways in mind are crucial to ensuring safe transport of hazardous materials. Preparation and diligence greatly mitigate the inherent risks found along this rural-urban connecting corridor.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	Medium	11.45352, 77.85698	30 KM/Hr
5	Turn	Medium	11.46048, 77.85036	30 KM/Hr
6	Turn	Medium	11.46303, 77.84945	30 KM/Hr
7	Turn	Medium	11.46318, 77.84913	30 KM/Hr
8	Turn	High	11.45542, 77.81725	15 KM/Hr
9	Turn	High	11.45634, 77.81673	15 KM/Hr
10	Turn	Medium	11.49421, 77.88021	30 KM/Hr
11	Turn	High	11.49383, 77.88526	15 KM/Hr
12	Turn	High	11.49395, 77.88537	15 KM/Hr
13	Turn	High	11.49409, 77.88536	15 KM/Hr
14	Turn	Medium	11.49435, 77.88499	30 KM/Hr
15	Turn	Medium	11.49465, 77.88488	30 KM/Hr
16	Turn	Medium	11.56072, 77.89919	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Turn	Medium	11.56096, 77.89931	30 KM/Hr
18	Turn	High	11.57035, 77.90223	15 KM/Hr
19	Turn	Medium	11.57041, 77.90221	30 KM/Hr
20	Turn	Medium	11.58279, 77.90658	30 KM/Hr
21	Turn	Medium	11.64197, 77.95197	30 KM/Hr
22	Blind Spot	Blind Spot	11.69590, 77.97114	10 KM/Hr
23	Turn	High	11.70133, 77.96363	15 KM/Hr
24	Turn	Medium	11.73427, 77.96723	30 KM/Hr
25	Turn	High	11.77772, 77.99628	15 KM/Hr
26	Blind Spot	Blind Spot	11.78535, 77.98924	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
3	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Veda Maruthuvamanai Hospital	11.6956155, 77.9678122	30 km/h	Medium
6	clinic	E.V.P Vairavel Clinic	11.6954975, 77.9667903	30 km/h	Medium
7	clinic	AMS Clinic	11.752664, 77.974974	30 km/h	Medium
8	hospital	Tholasampatti, Government Hospital	11.75638, 77.973813	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49421, 77.88021

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88526

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49395, 77.88537

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49409, 77.88536

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49435, 77.88499

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49465, 77.88488

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56072, 77.89919

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56096, 77.89931

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57035, 77.90223

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57041, 77.90221

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58279, 77.90658

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64197, 77.95197

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.69590, 77.97114

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70133, 77.96363

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73427, 77.96723

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.77772, 77.99628

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.78535, 77.98924